AutoDrive: Turn-Boundary Continuation Control for Long-Horizon Coding Agents

Anonymous Author(s)
Anonymous Institution
Anonymous City, Anonymous Country

Abstract

Coding agents can end a conversational turn while a safe, authorized next action remains. We call this event *premature session handoff*. We present AutoDrive, a session-scoped controller that chooses among Continue, Stop, and Defer and durably admits bounded follow-up work. The design separates model execution from continuation admission and reconciles deterministic input identifiers after a crash, while preserving user priority. The frozen study specifies 48 SWE-EVO tasks, a 384-trajectory policy and worker matrix, and five boundary-classification conditions. This is a pre-results manuscript: effectiveness estimates remain pending. Its six-step worker horizon induces terminal summaries, and its model-panel labels are a reference standard rather than human ground truth. Accordingly, the current design can test bounded continuation utility and agreement with that reference; natural handoff prevalence, human effort reduction, and runtime safety require additional evidence. We specify those limitations and the prospective extensions needed for journal-level conclusions.

CCS Concepts

• **Software and its engineering** → **Software creation and management**; • **Computing methodologies** → *Machine learning*.

Keywords

coding agents, autonomous agents, continuation, termination, abstention, long-horizon software engineering, fault tolerance

Draft status. The implementation and v1.14 protocol are available. Four v1.13 paid canaries and one accepted negative v1.14 non-primary pilot are descriptive only. The formal matrix and disclosed model-panel reference labels are incomplete. Every red Pending field is generated from an explicit result macro and is not an empirical claim.

1 Introduction

Interactive coding agents combine a language model, repository tools, execution feedback, and a conversational interface. Their provider turns are necessarily finite, but the user's software task may span many such turns. A useful distinction follows: a *worker turn* may end because the model emitted a terminal response, while the *admitted task* remains incomplete. In practice, a worker may summarize partial progress, list unexecuted next steps, or ask whether it should continue. The human then sends a semantically thin message—often only "continue"—that restores execution without adding requirements, authority, or information.

This interaction is not merely inconvenient. It fragments long-horizon execution, makes completion dependent on user vigilance, and confounds evaluation: a capable worker can receive a failing score because its interface yielded control at an arbitrary turn boundary. At the same time, automatically resuming every terminal response is unsafe. A worker may be waiting for a product choice, credentials, permission expansion, destructive confirmation, or missing acceptance criteria. Blind continuation can also create redundant model calls, unbounded loops, cost exhaustion, or repeated external effects.

Prior systems establish important neighboring ideas. ReAct interleaves reasoning and action [16]; coding scaffolds such as SWE-agent and OpenHands provide interfaces and execution environments [14, 15]; and Magentic-One uses an orchestrator to track progress and recover from errors [4]. SCATE directly addresses premature coding-agent termination through adaptive supervision for test generation [5]. Sequential abstention [11], evidence-linked completion [10], and infinite agentic loops [7] further constrain novelty. Our narrower question is how a terminal-turn continuation decision can be integrated with durable admission, user preemption, and recovery, and whether that integration offers measurable utility under controlled budgets.

We study that question through AutoDrive. The controller is scoped to the worker's repository location and the durable Session. It evaluates only safe turn boundaries and returns one of three actions. Continue admits one bounded queued input; Stop closes the automatic chain; and Defer returns control when the next action depends on human authority or information. A continuation decision is persisted before queue admission with a deterministic input identifier. Recovery can therefore reconcile a recorded decision with an existing or missing input without duplicating it.

This paper makes four bounded contributions:

(1) It specifies three terminal-boundary actions and the distinction between continuation permission and task-completion evidence.
(2) It integrates bounded continuation with durable Session admission, user-priority scheduling, and explicit crash-reconciliation invariants.
(3) It provides a frozen evaluation specification: 48 SWE-EVO tasks, four policies, three worker models, 384 planned trajectories, a 180-boundary target, five information conditions, and nine fault scenarios.
(4) It provides an artifact-validation and analysis harness, with explicit limits on which scientific claims its current measurements can support.

We do not claim that supervisors, memory, termination critics, abstention, or loop bounds are new. The empirical contribution is conditional on the frozen study and will be reported even if AutoDrive produces null or negative utility.

2 Problem Formulation

2.1 Sessions, turns, and boundaries

Let a Session be an admitted goal g, a durable history H_t, a repository location ℓ, a worker policy π_w, and a set of permissions P. A provider turn transforms history and environment state until it emits a terminal finish reason. We call that point a *turn boundary* b_t. Tool-internal continuations needed to settle already emitted calls are part of the same worker turn and are not AutoDrive decisions.

The latent task state at a boundary is one of three operational classes:

$$y_t \in \{C, S, D\},$$

where C means that a safe, concrete, in-scope action remains; S means that the goal is complete or no useful continuation is warranted; and D means that continuing requires a subjective choice, missing information, expanded authority, or a dangerous/external action. A boundary controller π_c predicts

$$\hat{y}_t = \pi_c(g, H_t, \ell, P, m_t),$$

where m_t is optional within-session controller memory. We call a boundary a *premature session handoff* when $y_t = C$ but the interface yields to the user rather than admitting an equivalent safe continuation.

2.2 Safety asymmetry

The error costs are asymmetric. Predicting C for S wastes work and can cause repeated effects; predicting C for D can bypass human control. Predicting S or D for C preserves safety but forfeits autonomy. Therefore, AutoDrive gives Defer precedence whenever authority or information is uncertain, and bounds each automatic chain even when all decisions are Continue.

We measure task utility and boundary classification separately. A controller can classify boundaries accurately yet fail to improve task resolution because the worker cannot solve the remaining problem. Conversely, blind continuation can improve resolution while producing unacceptable unsafe or redundant turns. No single success rate captures both dimensions.

2.3 Research questions

The study asks:

RQ1 How often do long-horizon coding trajectories contain premature handoff boundaries?

RQ2 How does continuation control affect resolution, partial progress, and manual continuation demand?

RQ3 How do model supervision, the deployed initial-goal and within-session-memory inputs, and an offline derived trajectory summary affect boundary decisions?

RQ4 What safety, redundancy, token, cost, latency, and recovery trade-offs follow?

3 System Design

3.1 Design overview

Figure 1 shows the control path. The worker owns model/tool execution. The controller runs only after the worker and pending user work reach a safe boundary. Its decision is not an in-memory loop flag; it becomes durable Session state. A Continue decision carries a chain identifier, continuation number, deterministic input identifier, reason, and next prompt. The input is then admitted using the same queue abstraction as human follow-ups.

3.2 Location and Session scope

Repository tools, permissions, model resolution, and filesystem context are location-scoped. Accordingly, the controller is resolved from the same Location service graph as the Session runner rather than from a process-global policy singleton. Its settings are Session state with the following precedence: Session override, project configuration, then built-in defaults. The built-in state is disabled, supervisor policy, at most five continuations, contextual prompting off, and within-session database memory on. A legacy boolean supervisor setting maps to the new policy without silently changing existing configurations.

The public V2 endpoint updates the Session state used by the UI, settings dialog, and slash command. A legacy V1 server rejects the operation rather than presenting a local toggle that did not change runtime behavior.

3.3 Three-state decision

The deterministic policy examines the trailing worker response for completion, continuation, choice, missing-information, permission-expansion, and destructive-action cues. Safety cues produce Defer before continuation patterns are considered. The supervisor policy receives the admitted initial goal, last worker output, optional trajectory context, and optional within-session memory. It must return strict JSON with an action, reason, optional next prompt, and optional memory update. Its tool set is empty, temperature is zero, output is capped at 1,024 tokens, and the call has a 15-second timeout.

Malformed output, provider failure, an unavailable model or Session, an empty response, or timeout produces Defer under protocol v1.14. The controller therefore abstains instead of converting uncertainty into a regex guess or an automatic continuation. Traces record the failure path separately from a semantic supervisor decision. Project playbooks are read only when explicitly configured. The default memory lives in the Session database and never writes `.opencode/auto-drive.md` into the worktree.

3.4 Safety policy and action semantics

The controller does not grant authority. It evaluates whether another worker turn may proceed under the authority already attached to the Session. Table 1 makes this distinction operational. A next step can be technically feasible yet still require Defer because it changes product behavior, spends money, publishes data, messages a person, deploys to a shared environment, or crosses a

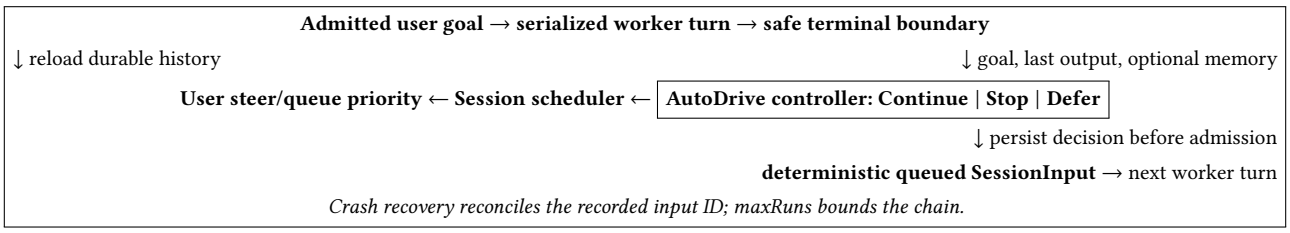


Figure 1: AutoDrive controls Session continuation at worker turn boundaries. Durable admission remains separate from model execution.

Table 1: Decision semantics. Defer takes precedence on uncertainty about authority or information.

Action	Necessary conditions and effect
Continue	Goal incomplete; concrete, in-scope, safe next step; admit one bounded queued input.
Stop	Completion is supported or no useful in-scope action remains; close the chain.
Defer	Choice, information, permission, or safety is unresolved; return control without admission.

destructive boundary. Conversely, a read-only diagnostic can remain Continue when it is a routine way to resolve uncertainty inside the admitted scope.

Let $A(b)$ denote the set of concrete next actions visible at boundary b, $\mathrm{scope}(a, P)$ whether action a is within the current permission set, and $\mathrm{safe}(a)$ whether it avoids a destructive or externally consequential transition. The deterministic admissibility condition is

$$\mathrm{admit}(b) = \neg\,\mathrm{complete}(b) \wedge \exists a \in A(b) : \mathrm{scope}(a, P) \wedge \mathrm{safe}(a) \wedge \neg\,\mathrm{choice}(a).$$

This predicate is not presented as a complete semantic model of software work. It documents the asymmetry implemented by the deterministic baseline and reference-label rubric: evidence of missing authority dominates evidence that work remains. The supervisor can supply a better interpretation of the trace, but cannot weaken this interface contract; failure to obtain that interpretation is itself a reason to defer.

3.5 Durable exactly-once admission

The critical crash window lies between deciding to continue and admitting its follow-up. AutoDrive closes it with the sequence below:

(1) Choose a deterministic input ID and persist an `auto-drive.decided` event containing the chain and continuation number.
(2) Admit a queued SessionInput with that exact ID and source metadata.
(3) On every drain start, reconcile any persisted Continue decision with the input store.
(4) If the input is absent, admit it. If an equivalent input exists, reuse it. If the same ID conflicts in Session, prompt, delivery, or source, fail closed.

This is exactly-once admission, not exactly-once provider execution. Provider work after a crash has no durable run identity in the current process-local scheduler; the paper does not claim clustered execution recovery. Duplicate advisory wakes coalesce at the Session coordinator, and durable inputs remain the source of truth.

Conditional admission argument. Assume that committed decision events survive restart, input IDs are unique in the durable store, admission compares the full input identity atomically, and a persisted decision always reuses its original ID. There can then be at most one admitted row for a decision: any second successful insertion would contradict ID uniqueness; an equivalent retry reuses the row and a conflicting retry fails. If the store becomes available and a recovery drain eventually runs, an unadmitted decision can also be admitted, provided it has not been superseded by user control. This establishes a conditional admission property, not model correctness, tool-side-effect idempotence, or a proof of the implementation. Process-crash tests must exercise the persistence assumptions independently.

Recovery is a reconciliation operation rather than a retry loop. For a persisted continuation decision $d = (s, c, n, i, p)$ with Session s, chain c, sequence number n, input ID i, and prompt p, the reconciler performs one of three transitions. If i is absent, it admits (i, p) and records the source metadata. If i names an equivalent pending input, it returns that row without mutation. If i names a non-equivalent input, it raises a conflict and admits nothing. Thus a crash can delay admission, but it cannot select a second ID or increment the chain twice.

The two persistence records serve different audit purposes. The updated event records settings changes and their effective source. The decided event records the controller action, reason, chain, count, prompt identity, policy path, and optional memory transition. The queued input repeats the chain and decision metadata so that transcript consumers can attribute an automatic user-like message without changing the schema of genuine user input.

3.6 Scheduling and bounds

Real user steering inputs are promoted before queued inputs. A user message begins a new automatic chain rather than inheriting a previous continuation count. Automatic follow-ups use queue delivery, so they do not interrupt an unsafe active point. Once the current drain would otherwise become idle, one queued input is promoted and the continuation condition is reevaluated.

Table 2: Frozen end-to-end trajectory matrix.

Worker setting	Tasks	Repeats	Trajectories
DeepSeek Pro primary	48	1	192
DeepSeek Pro subset repeats	12	2 extra	96
Qwen Max replication	12	1	48
DeepSeek Flash replication	12	1	48
Total			384

The persisted counter prevents a restart from resetting the limit. A request for continuation number $k + 1$ becomes STOP when k already equals `maxRuns`. Provider retry uses a finite policy; indefinite waiting is prohibited. Permission denial or user rejection interrupts the active ownership chain and leaves unrelated durable inputs intact.

4 Evaluation Methodology

4.1 Primary benchmark

SWE-bench established repository-level issue resolution as an agent benchmark [8]. SWE-EVO extends the setting to release-level software evolution. Its current public version contains 48 tasks from seven Python repositories, with an average patch spanning 21 files and validation averaging 874 tests [9]. We pin all 48 tasks to upstream commit `9b83d5af` and Arrow SHA-256 prefix `74e7c631`; the artifact records both full digests.

We do not use SWE-Bench Pro for the primary conclusion. An official audit estimated that roughly 30% of its public tasks are broken and identified overly strict tests, underspecified prompts, low coverage, and misleading prompts [13]. This choice does not imply that SWE-EVO is flawless; Section 10 treats benchmark validity as a separate threat.

4.2 Policies and run matrix

Table 2 freezes the paid trajectory count. The four policies are: an oracle that uses an external task verifier to inject a static continuation only while incomplete; blind continuation to the bound; the deterministic regex policy; and the full supervisor policy. The off condition is extracted from the first-boundary prefix of every paid trace and never triggers another model call.

The primary worker is `d-robotics/deepseek-v4-pro`. The twelve-task subset receives two extra primary-worker repeats and one run each with `d-robotics/qwen3.7-max` and `d-robotics/deepseek-v4-flash`. Every supervisor is `d-robotics/qwen3.8-max`, isolating one source of controller variation. Qwen provides a different-family replication, whereas DeepSeek Flash is only a same-family scale or variant replication. The subset covers every repository and samples low/high upstream PR and `FAIL_TO_PASS` proxies. Gateway model aliases are recorded as such and are not represented as direct-provider revisions.

Each run uses a pinned image digest and base commit, temperature zero, six worker steps per segment, five automatic continuations, 45 minutes, and at most two concurrent tasks. Exact provider model versions and normalized request hashes are mandatory trajectory fields. The host executor holds provider credentials; task containers do not receive them.

The sixth-step runtime prompt explicitly requests a summary and remaining work while disabling tools. This creates a controlled interruption opportunity. Consequently, the source study measures continuation need under this six-step interface, not the prevalence of spontaneous handoffs in unrestricted agents. The first-boundary prefix is an early-stopping comparator with a smaller execution allowance; an improvement over it alone cannot attribute benefit to an intelligent controller rather than additional computation. Blind continuation provides a bounded additional-computation reference, but realized token use still differs. Matched-budget and natural-boundary studies require a separately frozen extension.

Before either corpus collection or formal dispatch, a separately budgeted non-primary pilot must pass on SWE-bench Verified task `psf__requests-1142`. The pilot pins dataset revision `c104f840`, the Parquet and task-input hashes, the official harness commit, and the public AMD64 image digest. It writes only to an isolated pilot index and never contributes to an RQ estimate. One v1.14 run was accepted as a negative pilot: it remained unresolved with F2P 0/1, P2P 5/5, Fix Rate zero, no automatic continuation, 20,014 total tokens, and USD 0.121062 cost. Formal dispatch remains separately gated.

4.3 Execution and isolation

The harness treats the host executor as a narrow trust boundary. Before dispatch it reserves the run's maximum category charge and passes a frozen run object containing the task, worker, controller, strategy, repeat, timeout, and expected repository provenance. The executor creates the pinned task container without provider credentials. Model requests leave through the host; repository commands run inside the task container. The returned record is accepted only when its run and attempt identities match the reservation and all mandatory provenance fields parse.

The artifact set preserves every provider request and response, tool invocation, boundary snapshot, queue admission, promotion, verifier call, and cost event; controller outputs are preserved in the content-addressed provider-response records rather than duplicated into the Session trace. Before the worker prompt, the executor verifies the frozen HEAD and clean tracked state, then records pre-existing untracked files as a content-addressed manifest, binary patch, and Git tree. Boundary and final patches are computed relative to that tree. Updates below startup-only roots are omitted from the model patch but retained as trace provenance, so a charged trajectory remains graded and counted. Trajectory schema v4 makes this baseline evidence mandatory for boundary-source and formal acceptance. Raw traces are content-addressed, while the public index contains only their paths and hashes. This separation supports independent integrity checks without placing credentials or unrelated host state into the artifact. Secret-shaped strings cause rejection before append. The task container is discarded after grading, so memory and filesystem changes cannot cross task identities.

The oracle policy does not inspect a hidden reference patch. It invokes the same external task verifier used for final grading at a worker boundary and injects the frozen continuation string only when that verifier reports incomplete. Oracle therefore estimates the opportunity created by better boundary timing under an expensive verifier; it is not an upper bound on worker capability.

Blind, regex, and supervisor policies do not receive intermediate test labels from this verifier.

4.4 Boundary corpus

The source sampling frame is independent of the 384 formal rows: 96 supervisor-only trajectories, comprising all 48 SWE-EVO tasks with two repeats. A dedicated source marker in the deterministic run key prevents identity overlap. A boundary-scope capacity receipt, at most two concurrent tasks, an unchanged USD 102 source subcap, and an external `boundary/` index are mandatory. Source outcomes cannot enter RQ2 or the formal cost ledger.

The target is 180 real worker boundaries, balanced as 60 per class. Base trajectories are grouped into 54 development and 126 frozen test items. Extraction verifies artifact hashes and removes hidden reasoning and the supervisor's decision. The original reference panel comprises Qwen 3.7 Max and DeepSeek V4 Flash, with DeepSeek V4 Pro as adjudicator. The recorded availability amendment substitutes Qwen 3.8 Max when Flash coverage is insufficient, and GLM 5.3 when Pro coverage is insufficient. Requests, responses, usage, and exclusions remain auditable. The seal uses the schema label `independent-model-panel`; independence here means separately issued blinded requests, not independent model errors. In particular, the fallback pair shares a model family, and Qwen 3.8 is also the evaluated controller. Agreement with this panel cannot establish unbiased decision accuracy. Cohen's $\kappa \geq 0.75$ [2] is a freeze gate, not proof of label correctness. Failure of coverage, agreement, or balance leaves RQ3 unavailable under the current protocol.

The implemented offline conditions are: production regex; supervisor with last output only; last output plus initial goal; last output plus initial goal and a derived trajectory summary; and last output plus initial goal and within-session memory. Summary and memory are parallel additions to the goal baseline, as specified by the dated August 31 execution seal. The deployed v1.14 supervisor receives the information sources in the final condition, but the offline runner allows 60 seconds and rejects invalid outputs whereas deployment defers after 15 seconds or parse failure. Offline macro-F1 therefore measures agreement conditional on the offline response contract; it is not deployed-controller reliability. The October 3 erratum records a contradictory cumulative-input entry in the historical ablation JSON and identifies the execution seal and unchanged prompt builder as the operative specification. The historical file is retained; an archived run must identify its actual prompt inputs, not infer them from that stale entry.

Grouping by trajectory prevents adjacent boundaries from crossing the split, but does not prevent different repeats of the same SWE-EVO task from doing so. Any interpretation as held-out-task generalization is therefore unsupported. Later boundaries also depend on earlier source-controller choices. Memory deletion on a fixed trace estimates an information effect on those reachable states, not the end-to-end effect of disabling memory. A prospective extension needs task-level holdouts, independent human reference labels, and online memory-off trajectories.

Boundary sampling occurs after source trajectories are collected and before controller evaluation or formal dispatch. Candidates contain the admitted goal, terminal worker output, derived trajectory summary, patch, optional Session memory, and stable task and trajectory identifiers. Explicit permission state is not a separate candidate field; a judge can infer authority only to the extent that the captured text documents it. That limitation is especially important for DEFER labels. A label-blind contingency was sealed after an interim artifact-count check: if the completed 96-run frame contains fewer than 180 extractable boundaries, every task receives one disjoint repeat-2 source run; otherwise no augmentation runs. Supplemental collection cannot stop based on labels, agreement, or later yield. Sampling is then balanced to 60 adjudicated examples per class, including possible disagreements between the two initial judges; balancing does not guarantee agreement or estimate population frequency. RQ1 uses only the original unbalanced source frame, while RQ3 may use the augmented pool. Conflating these two estimands would bias the reported handoff frequency toward one third.

4.5 Fault injection

Nine scenarios target both correctness and safety: crash after decision/before admission; crash after admission/before promotion; duplicate wake; user input preemption; supervisor timeout; invalid supervisor JSON; retryable provider error; non-retryable provider error; and maximum continuation count. An accepted recovery must preserve input identity, queue priority, chain counter, and trace classification. Infrastructure failures alone may be rerun once with identical parameters, and only when no provider charge occurred.

4.6 Measures and analysis

Operational resolved rate requires both a completed execution and a passing official task grade. A passing patch attached to a failed execution is retained as a raw grader observation, not counted as operational success. Fix Rate follows SWE-EVO's partial metric over `FAIL_TO_PASS` and `PASS_TO_PASS` tests [9]; it describes the observed patch even when execution fails. The automated executor records token use, cost, wall latency, and unchanged-patch turns. A model timeout or loop represented by a valid trajectory remains a failed outcome. Charged harness exclusions and undispatched rows require a separate disposition table; the current complete-matrix analyzer refuses them rather than imputing zero-cost failures.

A manual continuation would be a user-originated request for progress without additional requirements or authority. The automated executor has no human participant and records this field as zero for every policy. Those zeros cannot estimate saved human interventions; the corresponding manuscript field remains pending. Similarly, the implemented `redundantTurns` field counts unchanged patch hashes. We call it the *unchanged-patch proxy*, because diagnostics and verification can be useful without changing a patch. Semantic redundancy requires separately adjudicated tool evidence. Offline predictions of CONTINUE on reference STOP or DEFER items are label-conditioned false-continuation rates, not observed permission violations. Actual unauthorized execution requires a permission-aware action trace. Ordinary trajectory completion is not a fault-injection recovery measurement.

Paired binary outcomes use a two-sided exact McNemar test [12]. If b tasks resolve only under the baseline and c only under treatment, the two-sided exact value is $2\Pr[X \leq \min(b, c)]$ for $X \sim$

Binomial$(b + c, 0.5)$, capped at one. Concordant tasks do not enter this test.

Primary repeat-zero differences use 10,000 paired task bootstrap resamples with percentile 95% intervals [3]. A fixed analysis seed is part of the environment record. The four supervisor contrasts—own off prefix, oracle, blind, and regex—form the Holm-corrected binary-outcome family [6]; raw and adjusted values are released. The extra repeats and twelve-task replications are descriptive, not additional independent tasks. Task resampling quantifies variation across the observed tasks, not across a population of independent repositories; repository sensitivity is needed because 26 of 48 tasks come from DVC. Failed valid trajectories remain in the analysis. Any missing required row prevents a complete-matrix result and is disclosed with its reason.

For RQ2, the primary estimand is the task-averaged supervisor-minus-off difference on the 48-task primary run. Regex, blind, and oracle contrasts are secondary members of the corrected family. For RQ3, the estimand is macro-F1 on the 126 frozen boundary items, accompanied by classwise confusion counts. For RQ4, false-continuation rates use label-conditioned denominators. Current paired cost and latency estimates require the complete accepted matrix; an all-dispatch cost estimate additionally needs charged exclusions and cannot be inferred from those pairs alone. Cross-model estimates retain the same definitions but are explicitly descriptive because the twelve-task subset is not powered as a replacement primary sample.

4.7 Budget and artifact gates

The hard cap is USD 800. The prospective budget amendment retains category ceilings of USD 50 pilot, 360 primary, and 288 cross-model, keeps the completed source-acquisition subcap at USD 102, and raises only post-source boundary research to USD 180. Before dispatch, the runner reserves no more than USD 1.25 per primary trace, USD 3.00 per cross-model trace, or USD 102/96 per boundary-source trace. Formal reservations include the immutable boundary ledger and the conservative prior-cost ledger, so the USD 800 hard total applies across stages even though their append-only ledgers remain separate. Each accepted trajectory includes the resolved model, request hash, image digest, code commit, trace hash, tokens, and cost. Secret-shaped material fails validation.

Charged harness failures that block a valid trajectory are sealed as exclusions: terminal usage, stable spend, the original receipt, and artifact hashes remain in the category ledger, but the record cannot enter sampling or RQ analysis and receives no retry. Actual spend is never truncated to the reserved amount. If lagged settlement reveals that a run exceeded its frozen per-run ceiling, the complete charge is recorded as a budget-overrun exclusion before the category cap is rechecked. Only a strict zero-provider infrastructure failure can consume the one allowed identical retry.

5 RQ1: Premature Handoff Frequency

The available-case estimate is the fraction of labeled first boundaries assigned Continue by the sealed model reference: Pending. Its denominator is Pending labeled boundaries from Pending planned source trajectories, with Pending missing. It concerns the primary worker under the imposed six-step limit. Missing labels or boundaries may depend on task difficulty and provider failures, so this conditional rate is not an all-source or natural-handoff estimate. If c observed first boundaries are CONTINUE, m are missing, and N were planned, the unidentified all-source proportion lies between c/N and $(c+m)/N$ when every missing item is allowed either class. These are worst-case bounds, not confidence intervals.

We will examine whether handoffs concentrate near agent step limits, after summaries, after partial test execution, or after provider stop reasons. This analysis distinguishes interface termination from substantive inability. It is descriptive and does not reclassify benchmark outcomes.

Table 3: Primary SWE-EVO outcomes. Values remain pending until all accepted traces pass provenance checks.

Policy	Resolved	Fix Rate
Off prefix	Pending	Pending
Oracle	Pending	Pending
Blind	Pending	Pending
Regex	Pending	Pending
Supervisor	Pending	Pending

6 RQ2: Task Utility

Table 3 is generated from the frozen trajectory index. The main paired contrast is full supervisor versus each trace's first-boundary prefix, followed by supervisor versus regex and blind policies. Oracle estimates the continuation opportunity under the same cap, not an achievable autonomous controller.

Before the formal matrix, we completed a paid v1.13 execution canary on one DVC task. Table 4 reports all four accepted policy trajectories. The canary used the same D-Robotics primary worker and controller identifiers as the v1.14 formal protocol, but its historical v1.13 run IDs are not formal matrix rows. Every trajectory was unresolved at its own first boundary; oracle and blind ultimately resolved, while regex stopped immediately and the supervisor resumed once before a controller timeout triggered the then-frozen regex fallback. These rows are independent, unpaired trajectories, so they establish mechanism and failure cases rather than a treatment effect.

The supervisor-minus-own-prefix resolved-rate difference is Pending, with task-bootstrap 95% interval Pending and Holm-adjusted p = Pending. The separately paired supervisor-minus-regex difference is Pending (95% interval Pending). We will not interpret a non-significant result as equivalence. Neither contrast measures saved human time or isolates controller quality from realized compute; the latter requires matched-budget evidence.

7 RQ3: Component Contributions

The frozen boundary macro-F1 for the deployed-input supervisor row is Pending. The offline table will show whether the initial goal resolves ambiguous "done" statements and compare a derived trajectory summary with within-session memory as parallel additions. We report the exact prompt composition for each row and

Table 4: Single-task v1.13 paid canary ablation on `iterative__dvc_2.21.1_2.21.2`. The four rows are independent, unpaired trajectories and are descriptive only. First and Final are binary resolution outcomes; Gain is the Fix Rate change from the trajectory's own first boundary.

Policy	First	Final	Gain	Cont.	Redun.	Worker	Ctrl.	Tokens	USD	Sec.
Oracle	0	1	1	5	4	36	0	505675	0.5665	1173.1
Blind	0	1	1	5	4	34	0	438262	0.4757	1040.6
Regex	0	0	0	0	0	6	0	23889	0.0484	436.1
Supervisor	0	0	0	1	1	12	2	96177	0.2837	532.9

per-class confusion matrices because macro-F1 can hide unsafe directionality.

Memory is evaluated within Session only. No default worktree file is written, and no information transfers across benchmark tasks. A fixed-history difference estimates the contribution of the supplied memory text on source-controller-reachable states. It does not estimate the effect of memory on the subsequent worker trajectory or establish that memory is beneficial online.

8 RQ4: Safety, Cost, and Recovery

The offline false-continuation rate on reference STOP is Pending and on reference DEFER is Pending. Their class-conditioned denominators and model reference must accompany the estimates. A decision error does not by itself demonstrate an executed unsafe action; a runtime safety study must trace permission checks and resulting side effects.

The supervisor-minus-regex monetary cost difference is Pending and wall-latency difference is Pending. Cost includes controller calls; token and verifier costs require separate accounting. Process-crash recovery is Pending until a distinct fault-campaign artifact records injected and recovered trials. The manuscript generator does not reuse the ordinary trajectory success flag for this quantity.

9 Error Analysis

After the frozen run, two fixed independent model reviewers assign one primary failure category and optional secondary categories to unresolved or unsafe cases, with the third fixed judge breaking disagreements. Planned categories are: worker capability failure; wrong continuation decision; correct continuation with ineffective next prompt; repeated strategy; stale or misleading summary; memory contamination within a Session; verifier mismatch; permission/defer error; provider/runtime failure; and benchmark defect. These are disclosed model judgments and are reported separately from executable grader outcomes.

The v1.13 canary supplies two descriptive cases. Regex persisted STOP on an empty patch while the worker listed unfinished implementation and validation and the F2P test failed. The supervisor durably admitted one continuation. Its next semantic response was CONTINUE, but it completed 16.839 seconds after release, beyond the 15-second deadline; the historical regex fallback persisted STOP. The extra segment produced no patch and increments the unchanged-patch proxy, which does not establish semantic redundancy. The case motivates inspecting latency and decision handling separately. Before any formal row was accepted, v1.14 changed failure handling to Defer while preserving the deadline and experimental factors. In Table 4, the historical column "Redun." denotes this patch-based proxy.

The accepted v1.14 non-primary pilot exposes the corresponding abstention path. Its controller produced valid Continue JSON 30.241 seconds after request release, but the 15-second deadline had already persisted Defer. No continuation occurred, the task remained unresolved, and all seven provider responses plus usage and settled cost were retained. We classify this as provider/runtime latency rather than a semantic policy error. It is a single descriptive pilot, not an ablation row or evidence of treatment effect.

During boundary acquisition, this rule excluded one harness failure in retry artifact routing. It had six complete responses, one proxy error, and USD 0.0906303 settled cost, but produced no trajectory or third attempt.

A later source run exposed a separate settlement and cost failure. All 40 sealed requests had successful, usage-complete responses, but three client retries rejected before provider admission were assigned later local sequence numbers. The old settlement predicate counted those unsealed errors and reported a false timeout. Immutable reconstruction verified 496,162 prompt and 14,355 completion tokens and USD 1.2186096 settled cost, which exceeded the frozen USD 102/96 run ceiling. We record the full charge as a non-retryable budget-overrun exclusion and admit neither a trajectory nor a boundary candidate. This is harness and budget evidence, not a policy outcome.

We will sample both false continuations and missed continuations, preserving task and worker strata. Error categories are descriptive and cannot remove a failure from quantitative analysis. Suspected benchmark defects are reported with evidence and sensitivity analyses, but the primary frozen result remains intact.

10 Threats to Validity

Construct validity. A static "continue" input is only an approximation to human follow-up; the automated study does not measure human effort. Its explicit six-step summary instruction induces boundaries, so even complete source data cannot establish natural handoff prevalence. The three labels compress nuanced states. The model panel is blinded but is not expert human judgment: fallback judges overlap with the controller family and, for Qwen 3.8, the controller itself. Separate calls and high kappa cannot eliminate that circularity. A task's repeats may also cross the trajectory-grouped split. We therefore restrict classifier findings to agreement with the stated reference on that split, not unbiased or held-out-task accuracy.

Internal validity. Provider services may be nondeterministic at temperature zero, and model aliases may move. We therefore record resolved model versions and request hashes, repeat the primary worker on a stratified subset, and pair each off prefix with its own trace. The same Qwen 3.8 Max controller across workers isolates one source of variation but may favor outputs it understands better. The gateway aliases do not reveal direct upstream revisions, limiting reproducibility outside the sealed gateway snapshot.

The single-task canary is especially vulnerable to this threat: its policy rows are independent trajectories, not counterfactual replays. We therefore report its costs, continuations, and outcomes descriptively and do not include it in McNemar tests, bootstrap intervals, or Holm correction.

The non-primary task image also contained untracked `build/` before worker execution. Consequently, the pilot's identical first and final build-only patch hashes preserve repository provenance but their byte and file counts cannot be attributed to the model. The official tests and unresolved classification remain usable. The schema-v4 baseline gate prevents recurrence by computing model patches from the recorded startup tree. Quarantining startup-root changes could hide intentional edits to generated artifacts, so every excluded path remains in the trace and must be reported in artifact diagnostics.

The gateway exposes cumulative spend only after settlement, so a long context can cross a per-run reservation before the host observes final cost. The harness records actual rather than capped spend and stops subsequent admissions when a ceiling is exhausted. Reservations cannot guarantee that delayed settlement never overshoots a cap. A charged excluded record cannot be recovered by treating it as zero cost or by silently rerunning it. Exclusion counts, reasons, and outcome bounds must accompany the sampling frame; absent a complete disposition analysis, complete-case estimates cannot stand in for the intended all-task result.

Benchmark validity. SWE-EVO is public, Python-only, and uneven across repositories; DVC contributes many tasks. Public histories may be contaminated in model training. The benchmark's tests can still be incomplete or implementation-specific. We publish task-level results and do not generalize to all software development.

External validity. The system is integrated into one agent runtime with durable Sessions. Agents without equivalent queue, event, or location abstractions may require different recovery mechanisms. The study covers coding tasks, not web purchase, messaging, trading, deployment, or other high-impact domains where Defer policies should be stricter.

Statistical conclusion validity. Forty-eight tasks limit power, and the twelve-task cross-model set is descriptive. The 384 trajectories do not constitute 384 independent task pairs. Repository concentration, repeated tasks, and grouped boundaries further reduce the independence available for generalization. We report intervals and corrected tests rather than using thresholded significance as the only evidence. Zero observed false continuations would still permit a nonzero population rate; safety cannot be asserted from a point estimate alone. Results are released even if effects are null, negative, or incomplete.

11 Related Work

Coding agents and benchmarks. SWE-bench evaluates issue resolution in real repositories [8]. SWE-agent shows that the agent-computer interface materially affects coding performance [15], while OpenHands provides a general platform with sandboxed execution and benchmark integration [14]. SWE-EVO expands the horizon from single issues to release-level changes and introduces partial Fix Rate [9]. AutoDrive changes neither worker capability nor task tests; it controls whether an admitted Session gets another bounded worker turn.

Orchestration and reflection. ReAct alternates reasoning with environment action [16]. Magentic-One's lead orchestrator plans and re-plans across a multi-agent team [4]. AutoDrive is narrower: a Location-scoped controller observes a single worker only at terminal turn boundaries and emits a durable Session scheduling decision. We therefore do not claim the first supervisor or progress tracker.

Premature termination and adaptive supervision. SCATE uses contextual-bandit supervision for coding-agent test generation, choosing interventions using coverage and cost feedback [5]. It is a close comparator, not merely background motivation. Its domain-specific state and reward cannot be transferred to release-level repair without a disclosed adaptation. AnalysisBench also reports premature termination and discrepancies between self-validation and independently checked outcomes [1]. Together, these works rule out novelty based only on identifying premature termination or automating a continuation request. AutoDrive's prospective distinction is the integration of defer-aware control with durable admission and user priority; its incremental value remains to be measured against resource-matched and close published controls.

Stopping and abstention. Agentic Abstention frames stopping under uncertainty as a sequential decision and shows that timing matters [11]. Evidence-Carrying Termination gates COMPLETE on typed, replayable evidence [10]. These works emphasize avoiding unsupported action or completion. AutoDrive's three states incorporate both concerns, but its central target is the complementary failure: returning control while safe work remains. It does not certify external truth or completion evidence.

Loop safety. Infinite Agentic Loops arise from unbounded feedback paths across model, tool, framework, and runtime behavior [7]. AutoDrive uses a persisted continuation limit, finite provider retries, and a Defer state. These are runtime bounds, not a static detector and not a proof that all loops are absent.

12 Conclusion

Provider turn completion and software task completion are different events. AutoDrive makes that distinction explicit with a three-state controller and durable, bounded follow-up admission. Its implementation and conditional admission argument motivate a controlled study, but do not establish an empirical gain. The frozen study concerns an imposed turn limit and a disclosed model reference. A separately specified journal extension must test natural boundaries, matched-resource controls, external reference judgments, and real process-crash recovery before broader claims are warranted. The current artifact is a pre-results research draft, not a completed journal evaluation.

Table 5: Mechanism-level positioning. Differences are design distinctions, not measured advantages or claims that other systems lack unlisted features.

Work	Relevant mechanism	Necessary comparison or scope boundary
SCATE [5]	Adaptive, coverage/cost-sensitive coding-agent supervision	Reproduce supported test-generation task or disclose a repair-domain adaptation.
Magentic-One [4]	Task and progress ledgers in team orchestration	Match worker resources for a ledger-controller comparison.
Agentic Abstention [11]	Sequential abstention and reusable stopping guidance	Distinguish fixed cross-episode rules from Session-local memory.
Evidence-Carrying Termination [10]	Completion gated by replayable trace evidence	Admission identity does not certify task completion or external truth.
AutoDrive (this work)	Three actions, bounded queue admission, crash reconciliation	Benefit beyond static extra compute and runtime safety remain empirical questions.

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A Frozen Protocol Details

A.1 Twelve-task replication subset

The subset is fixed before outcomes are observed:

- Conan: `2.0.2–2.0.3`, `2.0.14–2.0.15`;
- Dask: `2023.9.2–2023.9.3`, `2024.1.0–2024.1.1`;
- DVC: `0.30.0–0.30.1`, `3.13.3–3.14.0`, `2.8.1–2.8.2`;
- Modin: `0.24.0–0.24.1`, `0.25.0–0.25.1`;
- Requests: `v2.4.0–v2.4.1`;
- Pydantic: `v2.7.0–v2.7.1`;
- scikit-learn: `0.21.1–0.21.2`.

The selection covers all seven repositories and intentionally includes contrasting upstream PR and `FAIL_TO_PASS` counts. It is not optimized against AutoDrive outcomes.

A.2 Dataset composition

Table 6 summarizes the frozen manifest before any outcome is observed. Counts are descriptive properties copied from the pinned SWE-EVO Arrow artifact. They are included to expose the concentration of tasks: DVC supplies more than half of the instances, whereas Conan and scikit-learn supply two each. Primary estimates therefore report both micro averages and task-paired uncertainty rather than treating repository breadth as uniform.

Every task record additionally pins the start and end version, base commit, environment-setup commit, and container image name. Before dispatch, the executor resolves the image to a digest and records that digest in the trajectory. A mutable image name alone is insufficient provenance. The manifest is accepted only when it contains 48 unique instance identifiers, its repository distribution matches the source artifact, and the Arrow file's SHA-256 matches the frozen value.

A.3 Boundary decision rubric

Continue requires an incomplete admitted goal, a concrete next action, existing scope and permission, no material missing preference, and a routine or reversible engineering action. Stop requires proportionate completion evidence or absence of useful in-scope work. Defer applies to subjective choices, credentials or missing requirements, expanded permission, destructive/external/monetary action, or uncertainty only the user can resolve. Defer takes precedence over Continue.

Table 6: Frozen SWE-EVO task composition. FTP and PTP denote total failing-to-passing and passing-to-passing test counts in the manifest.

Repository	Tasks	FTP	PTP	Upstream PRs
conan-io/conan	2	80	966	62
dask/dask	8	3,004	26,179	9
iterative/dvc	26	502	2,039	172
modin-project/modin	3	73	1,501	12
psf/requests	4	9	515	23
pydantic/pydantic	3	238	6,112	109
scikit-learn/scikit-learn	2	2	731	2
Total	48	3,908	38,043	389

The target corpus contains 60 items per label; failure to obtain it is a study outcome, not permission to synthesize examples. Separate label files include a confidence level, evidence sentence, and next action or missing information. No supervisor prediction is visible to annotators. Base trajectories cannot cross development and test sets, but task identities can recur through different trajectories. The split therefore does not establish held-out-task performance.

Candidate acquisition has its own 96-run plan: all 48 SWE-EVO tasks, two repeats, the primary worker, and the full supervisor. A literal `boundary-corpus` component in the deterministic run key makes every source identity disjoint from the formal matrix. A fresh boundary-scope receipt must attest capacity 96 for both logical worker and controller. The command writes trajectories and its append-only ledger below external `boundary/`, never below the formal result path, and applies the USD 102/96 per-run ceiling. Every accepted source trajectory remains in the released sampling frame even when it produces no selectable item.

A.4 Annotation instrument and freeze

Judge models receive a rendered packet, not the raw result index. Its fields are the admitted goal, terminal worker output, derived trajectory summary, repository patch, and within-session memory. The prompt does not supply a dedicated worker/model identity or permission-state field. Permission or tool evidence can be judged only when present in the captured text; this is weaker than a complete authority audit. Costs, downstream outcomes, the current supervisor prediction, other labels, and reference patches are withheld. Historical memory remains visible and can encode earlier controller reasoning, another reason the model-panel reference is not independent of the system under study.

For every item, the annotator records: label; high, medium, or low confidence; one evidence sentence; a concrete next action for Continue; or the missing decision/information for Defer. A Stop label must point to completion evidence or explain why no useful in-scope action remains. Blank rationales fail schema validation.

Two fixed judge-model label files are sealed by hash before agreement is calculated. Cohen's kappa is computed over the three nominal labels, without confidence weighting. The third fixed judge supplies a separate adjudication file. If $\kappa < 0.75$ or any adjudicated class has fewer than 60 candidates after the label-blind augmentation, the RQ3 gate fails; judge prompts are not revised after observing labels. The released artifact preserves model identities and versions, requests, responses, complete usage, pre-adjudication files, their hashes, the agreement report, and adjudicated labels. The resulting reference standard is explicitly model-based rather than human gold.

The exact 54/126 split is group-constrained by base trajectory. The routine enforces split sizes, the full corpus's 60-per-class totals, and no base trajectory in both pools. It does not guarantee 18/42 per class in development/test; realized split counts must accompany every class-conditioned metric. General development work must not alter the fixed model-panel prompts after labels are observed. The older annotation guide's instruction to relabel after low kappa conflicts with the later dated amendment; the amendment's stop-and-report rule governs this frozen study, as recorded in the October 3 erratum. Test labels are opened only after controller prompts, parsing, fallback, and ablation ordering are frozen.

A.5 Fault invariants

A.6 Recovery procedure

The required process-crash campaign operates at named storage boundaries rather than arbitrary wall-clock delays; it is not yet a measured recovery-rate result. For the first crash window, a fail-point must terminate the process after the decided event commits and before input admission begins. The restarted drain must reload the event, recompute no decision, and reconcile the stored input ID. For the second window, termination occurs after admission commits but before the scheduler can promote the input. Restart must discover the pending row; creating a replacement is a failure even if the transcript text would match.

Duplicate wake testing issues multiple advisory wakes for the same Session while one drain is active. The process-local coordinator must join or coalesce them, while different Session IDs remain eligible for concurrent drains. User-preemption testing inserts both steering and queued user input around an automatic pending row and verifies real-user priority plus a fresh automatic chain. Timeout, provider-error, empty-response, unavailable-model, and invalid-JSON tests record the supervisor failure mode and require explicit DEFER. Provider-error tests also distinguish retryable from non-retryable worker errors and assert the five-attempt bound. The maximum-count scenario restarts between continuations to ensure the durable count, rather than process memory, enforces the limit.

Table 7: Frozen failure injection and required invariant.

Scenario	Required invariant
Decision before admission	Recovery admits the deterministic input exactly once.
Admission before promotion	The pending input survives restart and promotes once.
Duplicate wake	Within a live process, wakes coalesce; no duplicate SessionInput row. This is not cross-crash provider exactly-once execution.
User preemption	Real steer/queue input takes priority and begins a new chain.
Supervisor timeout	The call terminates at 15 seconds, persists DEFER, and admits no continuation.
Invalid JSON	Parsing persists DEFER without heuristic guessing or implicit continuation.
Retryable provider failure	Retries are finite and visible in the trace.
Non-retryable provider failure	Execution stops immediately and counts as failure.
Maximum count	No input numbered `maxRuns + 1` is admitted.

Recovery success at these admission failpoints requires one source input ID, correct delivery mode, unchanged prompt, correct chain and sequence metadata, no skipped user input, and a traceable terminal classification. Provider-call counts are checked within the tested pre-execution crash windows; they are not a guarantee of idempotent provider or tool effects after arbitrary interruption. Merely reaching an idle Session is not recovery evidence.

A.7 Artifact schema

Each trajectory index row contains four groups of fields:

- *identity*: run, task, policy, worker/controller model, repeat, and attempt;
- *outcome*: timestamps, status, failure class, resolution, Fix Rate, and first-boundary prefix outcome;
- *operation*: continuation, token, cost, latency, and unchanged-patch counters, plus historical fields for manual, safety, and recovery values that require independent measurement before interpretation; and
- *provenance*: resolved model version, normalized request SHA-256, image digest, task base and implementation commits, and content-addressed trace path.

A successful record cannot carry a failure class, and a failed record must carry one. A formal run ID must belong to the 384-row matrix; a boundary-source ID must belong to the separate 96-row plan. Either must match its task, worker, controller, strategy, and repeat. The request hash is computed after defaults are normalized but before credentials are attached. Trace hashes are checked before analysis so that replacing a raw trace invalidates the index.

The cost ledger is append-only. The validator rejects unknown run IDs, duplicate task matrix identities, category or total overrun, missing provenance, secret-shaped content, and analysis over an empty result set.

A.8 Budget reservation and failure accounting

The USD 800 cap is enforced before dispatch, not audited only after completion. The harness calculates current committed spend plus outstanding reservations. A primary run reserves at most USD 1.25, a cross-model run at most USD 3.00, and a boundary-source run at most USD 102/96; the post-source boundary-research cap is USD 180 after the prospective amendment. Every formal batch reads the immutable boundary and conservative prior-cost ledgers before reservation, while writing new costs only to the formal ledger. Dispatch is denied if either the applicable category or the global total cap would be crossed. Actual provider usage replaces the reservation when the record is appended. A mismatch between result index and ledger pauses subsequent execution for reconciliation.

Only a predefined, zero-charge infrastructure exit can consume the single rerun allowance. Examples include container creation failure before a request, corrupted task-image transfer before startup, or host interruption before provider dispatch. Model timeout, context exhaustion, controller loop, rate-limited provider attempts that incurred usage, verifier failure after execution, and budget exhaustion remain failed outcomes. This rule prevents selectively rerunning difficult or unfavorable trajectories.

If provider use occurs before a harness defect blocks a valid record, dispatch stops. Reconciliation requires one terminal event per request, complete usage for successful replies, stable spend samples, the original receipt, and verified artifact hashes. The exclusion consumes budget but cannot enter the empirical index or candidate pool and receives no retry. Released exclusion records keep missing rows auditable.

The USD 50 pilot pool is gated separately. Its frozen paid task is SWE-bench Verified `psf__requests-1142`, outside the 48 SWE-EVO instances. The manifest pins dataset revision `c104f840`, Parquet and task-input SHA-256 values, the official harness commit current at image creation, the `pytest -rA` test command, and AMD64 image digest prefix `sha256:66a8b693`. Its purpose is to validate container isolation, trace completeness, model identity capture, grader invocation, secret rejection, and cost reconciliation without selecting on a primary-benchmark outcome. Pilot output is isolated below `pilot/` and never enters RQ estimates. The accepted v1.14 r9 run was unresolved (F2P 0/1, P2P 5/5, Fix Rate zero), used 17,788 prompt and 2,226 completion tokens, and cost USD 0.121062. A valid controller continuation arrived after the deadline, so the Session persisted DEFER and performed no automatic continuation. Its image also exposed 65 pre-existing build paths; a later zero-network schema-v4 replay preserved them as baseline evidence and produced an empty model-relative patch.

An engineering preflight on 2026-08-30 deviated from the intended non-primary pilot by using the smallest locally feasible official image, task `iterative__dvc_0.30.0_0.30.1`, under a host disk constraint. No normal handoff boundary or code change was obtained, and the traces were rejected from the result index. The preflight exposed an invalid final-step model-role construction and a 20-request free-tier quota. Main dispatch therefore remains disabled. If the study proceeds, the unchanged all-48 estimate is accompanied by a sensitivity analysis excluding this canary-contaminated task.

After transport, request-routing, output-limit, controller-release, and grader amendments, the v1.13 D-Robotics canary completed one accepted trajectory for each of oracle, blind, regex, and supervisor policies on `iterative__dvc_2.21.1_2.21.2`. All four first boundaries were unresolved; oracle and blind eventually resolved, regex stopped immediately, and supervisor resumed once before a 15-second controller timeout invoked the historical v1.13 regex fallback. The four rows are retained only as a single-task, unpaired paid canary and do not enter RQ estimates. Before any formal row was accepted, v1.14 changed supervisor timeout, unavailable-model, provider-error, empty-response, and malformed-output handling to DEFER, retained the 15-second bound and model matrix, and regenerated all 384 run IDs. Formal dispatch remains disabled until the 96-run source collection, annotation freeze, and fresh full-capacity v1.14 receipt are accepted. The unchanged all-48 estimate will also be accompanied by the preregistered canary-contamination sensitivity analysis.

A.9 Model and continuation request contract

Table 8 lists the provider-independent request fields. Model aliases are planning identifiers only. At runtime the executor must capture the provider's resolved version and a SHA-256 over the normalized outbound request. A completed record with an unresolved version, missing request hash, nonzero temperature, or unrecognized default is rejected rather than treated as comparable.

All policies share the same worker system prompt, repository tool registry, permission set, per-segment step budget, context construction, and final verifier. Strategy-specific text appears only in the controller decision path and the common static continuation input. The supervisor has no repository tools and cannot mutate the task container directly. This prevents controller capability from becoming an unreported second coding agent.

The normalized request removes credentials and transport-only headers but retains provider, endpoint family, resolved model, messages, tools and schemas, sampling settings, token bounds, stop conditions, and provider feature flags. Array ordering is preserved. JSON object keys are canonicalized before hashing. If a provider rejects a frozen field or silently substitutes a model, execution pauses and the protocol records a deviation; the harness does not rewrite earlier hashes.

Automatic continuation uses queue delivery because a terminal worker response is not itself permission to interrupt an active tool boundary. The next prompt is deterministic for oracle and blind policies. Regex and supervisor may provide a concrete safe next prompt, but the stored decision and Session input must contain the same text. A genuine user steer or queue input wins scheduling priority and resets the automatic chain, even when it is semantically equivalent to the pending automatic prompt.

A.10 System acceptance beyond model outcomes

The implementation has three independent acceptance layers. First, unit and integration tests establish controller parsing, safety precedence, configuration precedence, persistence, reconciliation, scheduling, retry limits, and V1 rejection. Second, browser tests establish that new and adopted Sessions load their server-backed state, the status indicator follows decided events, settings and slash command update the same endpoint, and a legacy server cannot display a false success. Third, fault runs validate behavior across real process restarts and failpoints.

These layers must not be substituted for one another. A passing unit test does not estimate SWE-EVO utility; a successful task trajectory does not prove crash recovery; and a rendered UI toggle does not prove runtime state changed. The artifact checklist therefore keeps system, boundary, end-to-end, visual, and publication gates separate. The paper marks mechanism claims as test-backed only when the corresponding repository evidence exists, and leaves empirical result macros pending until the appropriate frozen data gate closes.

The public API is also versioned behavior. V2 accepts a Session-level update and returns effective state; Session information projects that state to clients. V1 returns an explicit unsupported response. Configuration resolution is Session override, project configuration, then built-in default. The legacy boolean supervisor option maps into the new policy vocabulary, while invalid continuation limits outside 1–20 fail validation. None of these compatibility paths may silently enable AutoDrive for an existing Session.

B Implementation Verification Map

Evidence map. Core decision tests cover tri-state parsing and safety priority; state/config tests cover precedence and legacy mapping; controller/projector integration covers persistence and exact admission; input promotion covers source metadata; Session runner tests cover user priority and concurrent resume; Server tests cover V2 and V1 behavior; Chromium tests cover controls; and the independent evaluation package covers matrices, statistics, budget, and provenance.

C Analysis and Release Procedure

C.1 Frozen snapshot construction

The release protocol requires exactly 384 unique accepted run IDs for complete-matrix inference, no unknown matrix identity, all trace hashes present, and reconciled spend. It additionally requires the run index, ledger, trace-hash list, label files, adjudication log, environment lock, model request manifest, and analysis commit to be hashed into one snapshot manifest. The complete snapshot is not yet available. Standalone analysis and result-rendering commands validate structure and numerical consistency; they do not independently bind every source artifact to a cryptographic seal. A successful command is therefore insufficient release evidence.

Table 8: Frozen request envelope. Provider-specific defaults must be resolved and hashed by the executor.

Component	Field	Frozen value
Worker	Models	D-Robotics DeepSeek V4 Pro; Qwen 3.7 Max; DeepSeek V4 Flash
Worker	Sampling	Temperature 0; low reasoning; 4,096 output tokens; tool choice auto
Worker	Horizon	Six steps per segment; five continuations; 2,700 seconds
Supervisor	Model and sampling	D-Robotics Qwen 3.8 Max; temperature 0; no tools
Supervisor	Bound	1,024 output tokens; 15 seconds; DEFER on failure
Continuation	Delivery	Queue; "Please proceed with the next step."
Off policy	Derivation	First-boundary prefix of the same paid trajectory

Derived release files must be written into a new directory named by the snapshot hash, never over raw JSONL.

Required release products are a run-level CSV, task-level paired CSV, policy summary JSON, boundary confusion matrices, raw and Holm-adjusted tests, cost ledger summary, failure taxonomy, and LaTeX result macros. Manuscript tables reference generated fragments. The current renderer requires the complete formal matrix and all five boundary summaries and refuses unsupported manual-effort and recovery values. It is a structural guard, not a substitute for label-seal verification or independent numerical review. Bootstrap settings and the generating analysis version must accompany each interval.

C.2 Numerical audit

An independent audit recomputes counts and means from the raw index, verifies that all rates have the documented denominator, and checks that off results are prefixes of their own paid trajectories. It compares each manuscript number with the generated source, recomputes exact tests using a second implementation, and spot-checks bootstrap pairing by task ID. Values are rounded only for presentation; statistical procedures consume unrounded data.

The audit also checks direction labels. A positive supervisor delta always means supervisor minus comparator; a manual-prompt reduction is reported with an explicit sign and unit; latency uses wall-clock milliseconds; and cost uses recorded USD rather than token-derived estimates when provider invoices disagree. Null, negative, and inconclusive results remain in the main table.

C.3 Anonymization and source packaging

The anonymous and signed manuscripts share the same body, bibliography, generated result files, and appendix. A single wrapper controls author metadata. Before public release, trace sanitization removes host paths, account identifiers, and unrelated repository secrets while retaining commands, outputs, model content, and hashes needed for scientific review. Redaction locations are logged; redaction does not change numerical inputs.

The arXiv source archive contains the placeholder-author wrapper, shared source, bibliography, appendix, and generated result fragments. It excludes raw traces, credentials, build caches, PDFs, and local absolute paths. Packaging alone does not certify scientific or submission readiness. Release verification requires unpacking into a clean directory and compiling with the same digest-locked, network-disabled TeX image. Author names, affiliations, email, order, license, disclosure text, and final PDF require separate user approval before any upload.

D Prospective Journal Extension (Not Executed)

The October 3 design document separates new experiments from the frozen v1.14 study. It is a proposal, not a new paid-run authorization or a source of empirical results. Its central question is whether defer-aware continuation improves externally verified task success at a fixed total resource envelope, compared with static continuation, an unsegmented native worker, and a relevant published controller.

Three studies address different identification problems. First, an unbalanced, outcome-blind natural-boundary sample distinguishes native terminal responses, imposed step limits, provider failures, permission gates, and exhausted budgets. Second, paired task experiments compare whole-session policies under common total worker-plus-controller budgets, with randomized blocked dispatch and a frozen model-epoch policy. Six-step and longer-segment conditions are separated from native execution; task-level and repository-sensitive analyses prevent repeated traces from becoming fictitious independent tasks. Third, external boundary judgments and real process-death fault tests evaluate authority decisions and durable admission independently of test resolution.

Fixed-trace ablations remove goal, summary, or memory information from an identical boundary, whereas online ablations disable one component for the full Session. The former measure decision dependence; only the latter measure downstream task effects. A three-state versus no-defer condition must be restricted to an isolated synthetic authority challenge with no real external effects. The complete protocol must specify its sample size, resource envelope, comparator adaptation, reference raters, inference family, missing-data rules, and immutable identifiers before collection. No positive result or sample size is a guarantee of acceptance.

E Result Publication Rules

All outcome macros are generated after the trajectory index contains 384 accepted run IDs and boundary labels pass the freeze gate. Tables and figures are regenerated from that snapshot. Hand-edited numerical results are prohibited. The final manuscript records the analysis commit, result snapshot hash, container digest, complete model identifiers, actual spend, and all deviations. The anonymous and signed PDFs are compiled from the same source with only author metadata changed.

Table 9: Frozen 48-task manifest. FTP and PTP are the task-level failing-to-passing and passing-to-passing test counts; PR is the upstream pull-request count.

Repository	Version evolution	FTP	PTP	PR
conan-io/conan	2.0.14–2.0.15	72	649	40
conan-io/conan	2.0.2–2.0.3	8	317	22
dask/dask	2022.9.2–2022.10.0	44	2,861	9
dask/dask	2023.3.2–2023.4.0	61	6,246	0
dask/dask	2023.6.0–2023.6.1	105	3,415	0
dask/dask	2023.6.1–2023.7.0	5	707	0
dask/dask	2023.8.0–2023.8.1	11	2,796	0
dask/dask	2023.9.2–2023.9.3	2	1,629	0
dask/dask	2024.1.0–2024.1.1	2,774	5,778	0
dask/dask	2024.3.1–2024.4.0	2	2,747	0
iterative/dvc	0.30.0–0.30.1	1	12	0
iterative/dvc	0.33.1–0.34.0	1	77	6
iterative/dvc	0.35.3–0.35.4	2	26	1
iterative/dvc	0.52.1–0.53.1	132	0	12
iterative/dvc	0.89.0–0.90.0	27	282	6
iterative/dvc	0.91.2–0.91.3	2	17	3
iterative/dvc	0.92.0–0.92.1	4	6	6
iterative/dvc	1.0.0a1–1.0.0a2	68	242	36
iterative/dvc	1.0.0b6–1.0.0	2	2	2
iterative/dvc	1.0.1–1.0.2	2	19	7
iterative/dvc	1.1.0–1.1.1	6	28	1
iterative/dvc	1.1.7–1.1.8	8	82	4
iterative/dvc	1.10.2–1.11.0	6	150	11
iterative/dvc	1.11.12–1.11.13	2	4	1
iterative/dvc	1.6.3–1.6.4	9	2	2
iterative/dvc	2.19.0–2.20.0	14	66	3
iterative/dvc	2.21.1–2.21.2	1	8	1
iterative/dvc	2.5.0–2.5.1	5	24	9
iterative/dvc	2.58.1–2.58.2	2	20	2
iterative/dvc	2.7.2–2.7.3	4	34	9
iterative/dvc	2.8.1–2.8.2	133	662	30
iterative/dvc	3.12.0–3.13.0	33	44	4
iterative/dvc	3.13.3–3.14.0	13	36	3
iterative/dvc	3.15.0–3.15.1	1	66	1
iterative/dvc	3.4.0–3.5.0	2	26	10
iterative/dvc	3.43.1–3.44.0	22	104	2
modin-project/modin	0.24.0–0.24.1	1	171	4
modin-project/modin	0.25.0–0.25.1	68	447	3
modin-project/modin	0.27.0–0.27.1	4	883	5
psf/requests	v2.12.2–v2.12.3	4	109	0
psf/requests	v2.27.0–v2.27.1	2	185	2
psf/requests	v2.4.0–v2.4.1	2	136	15
psf/requests	v2.9.0–v2.9.1	1	85	6
pydantic/pydantic	v2.6.0b1–v2.6.0	1	51	81
pydantic/pydantic	v2.7.0–v2.7.1	234	1,477	20
pydantic/pydantic	v2.7.1–v2.7.2	3	4,584	8
scikit-learn/scikit-learn	0.20.1–0.20.2	1	438	0
scikit-learn/scikit-learn	0.21.1–0.21.2	1	293	2